

SILENT ALIASING IN FIXED FOURIER-FEATURE COORDINATE MODELS

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ABSTRACT

Coordinate models built on a *fixed* finite set of Fourier features—the linear core of Fourier-feature implicit neural representations—fit signals by least squares on an exponential dictionary Λ . We give an identifiability theory for what such models do to out-of-band content. (i) On any subset of a rate- Q sampling grid, an integer tone congruent to a model frequency modulo Q is *exactly* indistinguishable from that atom: zero training residual, full-energy reconstruction failure, and—by a two-point argument—no estimator whatsoever can avoid the error (*silent aliasing*). (ii) Random sampling provably destroys this exact aliasing: under timing jitter the tone’s visibility follows the jitter’s characteristic function ($v \approx 2\pi|r|Q\sigma_t$ for small Gaussian jitter), and under i.i.d. sampling the worst-case aliasability over a finite candidate set concentrates at $O(\sqrt{m \log(mK/\delta)/N})$. (iii) Detection is dichotomous: exactly coherent folds are undetectable by any sample-only test (identical observation laws), while visible tones admit exact noncentral- χ^2 detection power, which calibrated sample-only detectors match empirically. Every closed form is validated against Monte Carlo; trained nonlinear networks (Fourier-feature MLPs, SIRENs) are studied strictly as empirical extrapolation, with fold patterns predicted by each network’s own NTK linearization and all seed-level failures reported.

Index Terms— Sampling theory, aliasing, identifiability, Fourier features, coordinate models.

1. INTRODUCTION

Coordinate networks with Fourier features are now standard continuous signal models [1, 2, 3]. Their analytically exact core—and the object of every theorem in this paper—is the *fixed Fourier-feature coordinate model*: a finite frequency set $\Lambda = \{\omega_1, \dots, \omega_m\} \subset \mathbb{R}$ with linear coefficients, fit by least squares. Trained nonlinear networks (Fourier-feature MLPs, SIRENs) enter only as *empirical extrapolation* in Sec. 4, and adaptive or learned frequency sets are outside the scope of our theory; we do not claim results for “all INRs”.

Questions. When a signal contains energy outside Λ , three practical questions arise. **Q1**: when is that energy *exactly* invisible in the samples—fitting perfectly while corrupting the reconstruction? **Q2**: how does the sampling design (grids, jitter, i.i.d. draws) change this? **Q3**: when can a practitioner *detect* the problem from the samples alone?

Contributions. (1) An identifiability framework—sample atoms, equivalence classes, *visibility* $v_T(\nu)$, *aliasability* $a_T(\nu)$ —in which exact silent aliasing is classified on arbitrary subsets of sampling grids, with an estimator-independent two-point lower bound (Thm. 1). (2) Function-space accounting: a closed-form continuous Gram matrix with Riesz bounds and an exact error decomposition keeping coefficient error, aliasing-induced modeled-component error, and

truncation error *separate* (Thm. 2). (3) Proof that *random sampling breaks exact aliasing*, with explicit rates: a characteristic-function law for jitter and a finite-candidate concentration bound for i.i.d. sampling (Thm. 3). (4) A detection dichotomy: impossibility for coherent folds, exact noncentral- χ^2 power otherwise, and a calibrated empirical operating characteristic with baselines (Thm. 4, Sec. 4).

Relation to prior work (per-item increments; full audit in the repository). Classical modulo- N aliasing [4] is the full-grid special case of Thm. 1, which covers *arbitrary* grid subsets. Generalized sampling [5, 6] supplies the least-squares recast we build on; the equivalence-class, sampling-design, and detection results are not part of it. The generalized aliasing decomposition of eGAD [7] computes an aliasing operator for *given* samples; our $a_T(\nu)$ is a column norm of that operator, and our new content is probabilistic control of it *over the sampling design* (Thm. 3). Basis mismatch [8] concerns tones *near* atoms; silent aliasing is the complementary far regime. The Lomb–Scargle spectral window [9] predicts periodogram leakage; we add the estimator-independent indistinguishability statement and detection theory. The structured-dictionary view of INRs [10] motivates the model class but contains no sampling theory; INR sampling theorems to date [11, 12, 13] are achievability results for realizable signals; an NTK bound from missing frequencies [14] gives an error floor but not the fold structure, invisibility, or detection. Empirical SIREN aliasing has been observed [15]; we supply theory for the linear core and a measured, calibrated diagnostic.

2. SETUP AND BACKGROUND

Signals live on $[0, 1]$; $e_\omega(x) = e^{i2\pi\omega x}$. The model class is $\mathcal{M}_\Lambda = \{\sum_k c_k e_{\omega_k}\}$ with Λ *fixed*. Samples $T = \{t_1, \dots, t_N\} \subset [0, 1]$ define $S_T f = (f(t_j))_j$, the synthesis matrix $\Phi \in \mathbb{C}^{N \times m}$, $\Phi_{jk} = e_{\omega_k}(t_j)$, and observations $\mathbf{y} = S_T f + \boldsymbol{\varepsilon}$. We write $f = f_{\text{in}} + f_{\text{out}}$ with $f_{\text{in}} = \sum_k c_k^* e_{\omega_k}$ and $f_{\text{out}} = \sum_l a_l e_{\nu_l}$, $\nu_l \notin \Lambda$. Real signals are fit with the equivalent real cosine/sine design (Hermitian symmetry enforced in the parameterization; see the code). Ordinary least squares $\hat{\mathbf{c}} = \Phi^\dagger \mathbf{y}$ is used throughout the theory (no ridge); the underdetermined minimum-norm case is treated separately in the code and experiments.

Lemma 1 (LS stability; standard). *If $\sigma_{\min}(\Phi) > 0$ and $f_{\text{out}} = 0$ then $\|\hat{\mathbf{c}} - \mathbf{c}^*\| \leq \|\boldsymbol{\varepsilon}\|/\sigma_{\min}(\Phi)$ and, for white noise, $\mathbb{E}\|\hat{\mathbf{c}} - \mathbf{c}^*\|^2 = \sigma^2 \text{tr}((\Phi^* \Phi)^{-1})$.*

This is textbook least-squares perturbation, stated for calibration of what follows; the average-case bias/variance decomposition under random out-of-band content is likewise standard bookkeeping and is deferred to the supplement (Prop. S1).

Code, data provenance, and the supplement with full proofs: github.com/appleweiping/inr-aliasing-limits.

3. MAIN RESULTS

3.1. T1: exact equivalence and silent aliasing

For $\nu \in \mathbb{R}$ define the *sample atom* $\phi_\nu = (e^{i2\pi\nu t_j})_j$, the *visibility* and *aliasability*

$$v_T(\nu) = \frac{1}{\sqrt{N}} \|(I - P_\Lambda)\phi_\nu\|, \quad a_T(\nu) = \|\Phi^\dagger \phi_\nu\|, \quad (1)$$

with P_Λ the orthogonal projector onto $\text{range } \Phi$. $v_T(\nu) \in [0, 1]$ is the fraction of the tone's sample energy the model cannot absorb; $a_T(\nu)$ is the coefficient bias a unit tone induces.

Theorem 1 (Exact equivalence on grid subsets). *Let $\Lambda \subset \mathbb{Z}$, let T be any subset of the rate- Q grid $\{0, \dots, Q-1\}/Q$ with $\sigma_{\min}(\Phi) > 0$ (which requires $N \geq m$ and the elements of Λ distinct mod Q), and let $\nu \in \mathbb{Z}$. (i) $v_T(\nu) = 0$ iff $\phi_\nu \in \text{range } \Phi$, i.e. iff a tone at ν is exactly reproducible by the model class on the samples. If $\nu \equiv \omega_k \pmod{Q}$ then $\phi_\nu = \phi_{\omega_k}$, hence $v_T(\nu) = 0$ for every such subset T , the fit of a unit tone is the unit coefficient on atom k , and the training residual is exactly zero. (ii) (estimator-independent converse) With $f_1 = f_{\text{in}} + a e_\nu$ and $f_2 = f_{\text{in}} + a e_{\omega_k}$, $\nu \equiv \omega_k \pmod{Q}$, $\nu \neq \omega_k$: $S_T f_1 = S_T f_2$, so under any noise law the two hypotheses induce identical observation distributions, and any estimator $\hat{f}$ satisfies $\max_{i \in \{1,2\}} \|\hat{f} - f_i\|_{L^2[0,1]} \geq |a|/\sqrt{2}$. (iii) If $\nu \not\equiv \omega_k \pmod{Q}$ for all k and $[\Phi \ \phi_\nu]$ has full column rank, then $v_T(\nu) > 0$.*

Sketch. (i) $e^{i2\pi\nu g/Q}$ depends on $\nu \pmod{Q}$; (ii) is a two-point (Le Cam) argument with $\|e_\nu - e_{\omega_k}\|_{L^2} = \sqrt{2}$; (iii) is the rank condition. Full proofs in the supplement. *Silent aliasing* is regime (i)–(ii): the model fits the samples to the noise floor while the reconstruction error is at least the full out-of-band energy, at any budget N satisfying the rank condition, with ν arbitrarily far from Λ . Classical aliasing is the special case $T = \text{full grid}$ ($Q = N$) [4]; the point here is that *nonuniform designs inherit the grid's equivalence classes* whenever they live on a grid.

3.2. T2: function-space accounting

Coefficient norms are not signal energies unless the dictionary is orthonormal. Let $G_{L^2}[k, \ell] = \langle e_{\omega_\ell}, e_{\omega_k} \rangle_{L^2[0,1]} = e^{i\pi\delta} \text{sinc}(\delta)$, $\delta = \omega_\ell - \omega_k$, $\text{sinc}(x) = \sin(\pi x)/(\pi x)$.

Theorem 2 (Riesz conversion and exact decomposition). (i) With $\lambda_{\min}, \lambda_{\max}$ the extreme eigenvalues of G_{L^2} (positive iff the frequencies are distinct), $\lambda_{\min} \|c\|^2 \leq \|\sum_k c_k e_{\omega_k}\|_{L^2}^2 \leq \lambda_{\max} \|c\|^2$. (ii) With $\Delta c = \hat{c} - c^*$,

$$\|\hat{f} - f\|_{L^2}^2 = \Delta c^* G_{\Lambda\Lambda} \Delta c - 2 \text{Re } \Delta c^* G_{\Lambda, \text{out}} a + a^* G_{\text{out}} a, \quad (2)$$

the three terms being the aliasing-induced modeled-component error, a cross term (zero for all-integer frequencies), and the truncation energy.

Equation (2) is elementary but load-bearing: truncation error exists for *any* sampling and is not aliasing; only the first term is the model *misplacing* energy, and only that term responds to sampling design (Fig. 1d).

3.3. T3: random sampling breaks exact aliasing

Theorem 3 (Sampling design). *Let $\Lambda \subset \mathbb{Z}$ with $|\Lambda| = m$. (a) (jitter) Let $t_j = g_j/Q + \eta_j$ with g_j grid points and η_j i.i.d., mean-zero, with characteristic function χ_η , and let $\nu = \omega_k + rQ$, $r \in \mathbb{Z} \setminus \{0\}$.*

Then the empirical fold coherence $\frac{1}{N} \sum_j e^{i2\pi(\nu - \omega_k)t_j}$ has mean $\chi_\eta(2\pi rQ)$ and concentrates around it at rate $2\sqrt{\log(4/\delta)/N}$; consequently $v_T(\nu)^2 = 1 - |\chi_\eta(2\pi rQ)|^2 + O_P(m\sqrt{\log(m/\delta)/N})$.

For Gaussian jitter of std σ_t , $\chi_\eta(2\pi rQ) = e^{-2(\pi rQ\sigma_t)^2}$, giving the small-jitter law $v_T(\nu) \approx 2\pi|T|Q\sigma_t$. (b) (i.i.d. samples) Let t_j be i.i.d. from a density whose limiting normalized Gram satisfies $G_\infty \succeq \lambda_{\min} \mathbf{I}$ and whose limiting model-candidate coherences vanish on a finite candidate set Ω , $|\Omega| = K$ (for the uniform density and integer-disjoint candidates, $G_\infty = \mathbf{I}$, $\lambda_{\min} = 1$). With $\varepsilon = 2\sqrt{\log(4(mK + m^2)/\delta)/N}$ and $m\varepsilon < \lambda_{\min}$, with probability at least $1 - \delta$,

$$\max_{\nu \in \Omega} a_T(\nu) \leq \frac{\sqrt{m}\varepsilon}{\lambda_{\min} - m\varepsilon}. \quad (3)$$

Sketch. (a) On the grid part the fold phase is exactly 1; the jitter contributes i.i.d. unit-modulus factors, so the coherence is an empirical characteristic function; Hoeffding on real and imaginary parts. (b) Hoeffding + union bound over all $mK + m^2$ coherences, then a perturbation bound on $a_T(\nu) = \|(\Phi^* \Phi/N)^{-1}(\Phi^* \phi_\nu/N)\|$. Proofs in the supplement; both laws are verified to tight tolerance in Fig. 1(a,b).

Three honest distinctions follow. *Exact* grid equivalence persists at every N if the design stays on the grid (Thm. 1); for a fixed separated tone, randomized designs make aliasing vanish at explicit rates (Thm. 3); and at any finite N an adversary may still pick near-invisible alternatives—but under i.i.d. sampling even the adversarial candidate within a finite set becomes visible (Fig. 1c). We make no claim that “no sampling density removes aliasing”; what no design removes is the truncation term of (2).

3.4. T4: detection—impossibility and power

Theorem 4 (Detection dichotomy). (a) If $S_T f_1 = S_T f_2$, then for any noise distribution the observation laws under the two hypotheses coincide, and every sample-based test has power equal to its size. (b) For $y = D\beta + s + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$, D the real design of rank p and s the sample vector of an additive component, the residual statistic $\|(I - P_D)y\|^2/\sigma^2$ is χ_{N-p}^2 under H_0 and noncentral $\chi_{N-p}^2(\lambda)$ with $\lambda = \|(I - P_D)s\|^2/\sigma^2$ under H_1 : detection power is driven by amplitude times visibility, and is exactly the size α when $\lambda = 0$ (coherent fold).

Sample-only detectors must be *calibrated*: in all experiments, thresholds come from an independent null calibration set and are then frozen; test-set FPR/TPR are reported separately (Sec. 4).

4. EXPERIMENTS

All identities and bounds above are validated against Monte Carlo in the released test suite (34 tests); experiment JSONs carry seeds, configs, and environment stamps.

Theorem validation (Fig. 1). (a) the jitter law: empirical visibility of a grid-coherent tone matches $\sqrt{1 - \chi_\eta^2}$ across two decades, with the exact fold ($\nu \approx 10^{-13}$) at $\sigma_t = 0$; (b) worst-case aliasability over a $K=31$ candidate set under i.i.d. sampling decays as $N^{-1/2}$ beneath the proved bound (loose, as unions are, but finite exactly where the theorem says); (c) persistence vs randomization: the grid-subset coherent tone stays exactly invisible at every N , while under i.i.d. sampling both the same tone and the per- N adversarial candidate are immediately visible; (d) the decomposition:

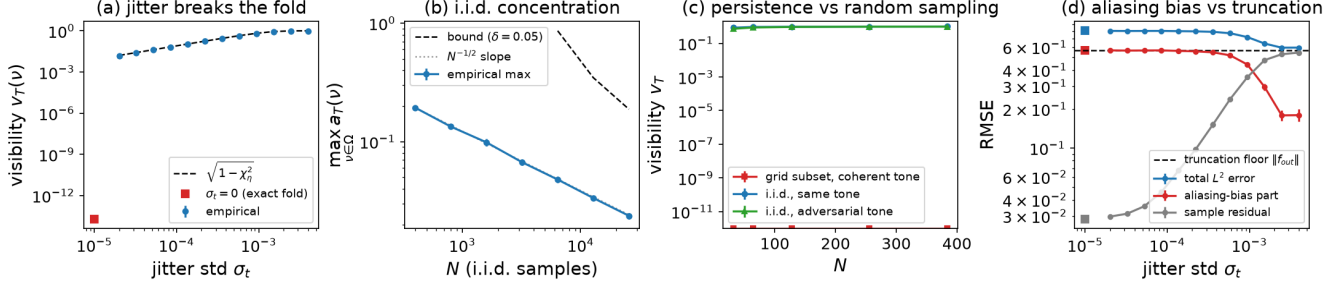


Fig. 1. Theorem validation (mean $\pm$ 95% CI over 20 draws). (a) T3a jitter law; (b) T3b concentration with the proved bound; (c) T1 grid persistence vs i.i.d. sampling (fixed and adversarial tones); (d) T2 decomposition: jitter removes aliasing bias, not truncation.

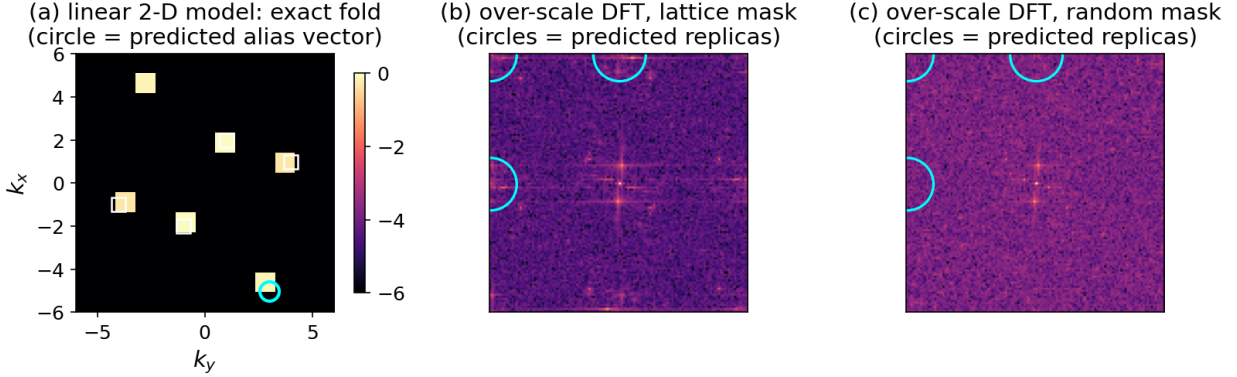


Fig. 2. 2-D structured aliasing. (a) linear model: the planted out-of-band frequency vector folds exactly onto the predicted atom (circle; the Hermitian partner appears mirrored) on a grid subset; (b,c) trained 2-D networks: DFT of over-scaled reconstructions shows the predicted replicas (circles) under a lattice (coherent) mask but not under a random mask of equal density.

jitter drives the aliasing-bias term to the noise level while the truncation floor is untouched—sampling design removes aliasing, never truncation.

Detection (Fig. 3). With calibration/test separation, sample-only detectors (ring test, residual, Lomb–Scargle band test, held-out prediction, cross-fit) track the exact Thm. 4(b) power curve; at the calibrated threshold the test-set FPR stays at its nominal 5%. On the exactly coherent grid fold every detector collapses (ring AUC 0.21, TPR at 5% FPR 0.03)—the indistinguishability of Thm. 4(a) made visible, reported separately from the operating characteristics by design.

Real signals, sample-only. On real series (Mauna Loa CO₂, sunspots, and nine 2-s speech segments from three Open Speech Repository recordings; anti-aliased `resample_poly` decimation; full records, no window selection), bandwidth selectors that see *only the training samples* are compared against a true oracle (the bandwidth minimizing the actual reconstruction error; evaluation-only upper bound). On CO₂ the sample-only Lomb–Scargle selector attains RMSE 0.167 vs the oracle’s 0.171 (means over 10 draws; paired CIs released)—deployable selection is not the bottleneck. On broadband speech *no* selector beats RMSE $\approx$ 1.00 (oracle 1.00, periodogram 1.46): at this sampling budget the truncation term of (2) dominates, the real-data face of the converse. Cross-validated selection is markedly less stable under gappy sampling. Full per-method, per-signal tables with paired 95% CIs are in the repository.

Trained networks (empirical extrapolation only). For FF-MLPs, SIRENs, and an explicitly band-limited head, tone attribution is

measured by an ablation (identical initialization trained with/without the tone) and compared to the prediction of *each network’s own NTK linearization*—no arbitrary reference band. Across ≥ 20 sampling seeds per condition, with label-permutation, amplitude, and weight-seed controls, the linear picture extrapolates well to *lazy-regime* networks—FF-MLP: exact fold match 77%, top-3 98%, pattern correlation median 0.96 ($n=60$); band-limited head: correlation median 0.98—but *fails* for SIRENs (exact match 20%, correlation median 0.49, $n=20$): trained SIRENs leave their initialization NTK, and we accordingly make *no* claim that the theory covers them (Fig. 4). In 2-D (Fig. 2), the linear model reproduces the predicted frequency-vector fold exactly ((a): planted (27, 3) on a rate-32 grid subset folds to (−5, 3), residual 10^{-15}), and trained 2-D networks show the predicted DFT replicas under *lattice* (coherent) masks—replica-energy ratio 40 ± 32 at the predicted shifts, vs 0.99 ± 0.09 under random masks of equal density (3 images, 12/24 mask draws)—while fitting the observed pixels to 66 dB against 12 dB on held-out pixels (2-D silent aliasing). Ridge and early stopping do *not* remove the replicas (ratios 89 and 25): structured aliasing is a sampling-design property, not an overfitting artifact; clean/noisy/held-out/full-field PSNR are reported separately in the released results.

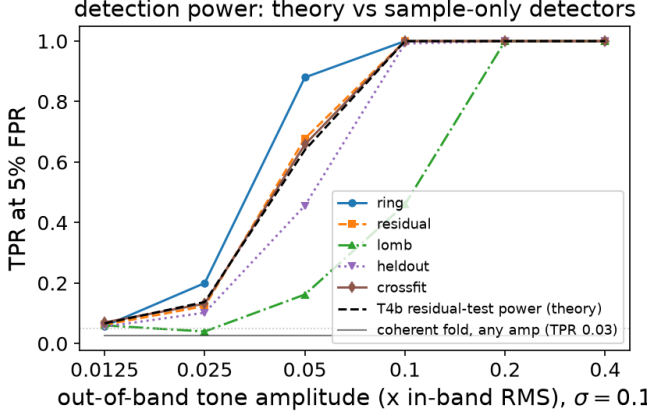


Fig. 3. Detection power (TPR at 5% FPR, calibration/test separated) vs tone amplitude at $\sigma=0.1$: five sample-only detectors against the exact Thm. 4(b) power curve; the coherent fold stays at chance at any amplitude (Thm. 4(a)).

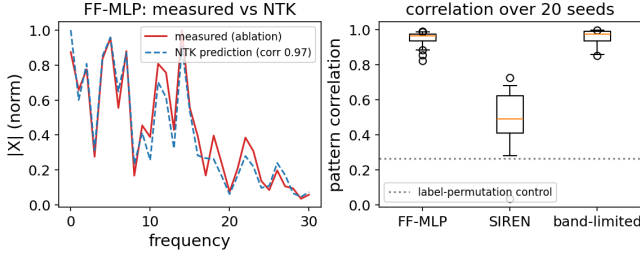


Fig. 4. Trained networks (empirical): ablation-measured tone attribution vs each network’s NTK prediction; right: pattern-correlation distributions over 20 sampling seeds (whiskers 5–95%), with the label-permutation control.

5. LIMITATIONS AND CONCLUSION

Our theorems cover fixed-feature, linear-coefficient Fourier models; trained nonlinear networks are studied only empirically—and the extrapolation has an empirical boundary: it holds for lazy-regime FF-MLPs but *fails for SIRENs*, whose training leaves the initialization NTK. Adaptive/learned frequency sets are open. The concentration bound is a union bound (loose constants); the jitter law’s multi-atom correction is $O_P(m\sqrt{\log/N})$. Within that scope the theory is complete and mechanically verified: exact silent aliasing is a property of *grid designs* and is classified by congruence; randomized sampling provably destroys it at explicit rates; detectability is dichotomous and quantitatively matched by calibrated sample-only detectors. Practical guidance: avoid pure grid designs when out-of-band content is possible; jitter of std σ_t buys visibility $\approx 2\pi|r|Q\sigma_t$; calibrate diagnostics on explicit nulls; and never read training residual as reconstruction fidelity.

6. REFERENCES

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